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How Can I Remove a Specific Item from an Array?



James W. — Updated: September 17, 2026

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The Problem

If you want to remove an item from an array, you can use the `pop()` method to remove the last element or the `shift()` method to remove the first element.

However, if the item you want to remove is not the first or last element, these methods are not the tools for the job.

The Solution

There are a few methods you can use to remove specific elements from an array in JavaScript. If the `pop()` or `shift()` methods won't work for your purposes, you can select a method depending on whether you'll identify the item by its value or its index.

Removing the Last Element of an Array

The `pop()` method removes and returns the last element of an array.

```
const myArray = [1, 2, 3, 4, 5];

const x = myArray.pop();

console.log(`myArray values: ${myArray}`);
console.log(`variable x value: ${x}`);
```

Output:

```
myArray values: 1,2,3,4
variable x value: 5
```

Removing the First Element of an Array

The `shift()` method removes and returns the first element of an array.

```
const myArray = [1, 2, 3, 4, 5];

const x = myArray.shift();

console.log(`myArray values: ${myArray}`);
console.log(`variable x value: ${x}`);
```

Output:

```
myArray values = 2,3,4,5
variable x value: 1
```

Removing an Element by Index

If you are identifying the item to be removed by its index, you can use the `delete` operator. If you want to use the value of the item you are removing, use the `splice()` method.

The `delete` Operator

The `delete` operator deletes the object property at the specified

index, but does not update the length of the array, and the value at that index of the array will be `undefined` .

```
const myArray = [1, 2, 3, 4, 5];

delete myArray[1];

console.log(`myArray values: ${myArray}`);
```

Output:

```
myArray values: 1,,3,4,5
```

The `splice()` Method

The `splice()` method takes two arguments, the index of the element you wish to remove and the index you wish to remove up to.

The `splice()` method creates a new array that stores all the values that were removed from the original array. The original array will no longer contain the values removed, and its length will be updated.

```
const myArray = [1, 2, 3, 4, 5];

const x = myArray.splice(1, 1);

console.log(`myArray values: ${myArray}`);
console.log(`variable x value: ${x}`);
```

Output:

```
myArray values: 1,3,4,5
variable x value: 2
```

Removing an Element by Value

If you are identifying the element to be removed by its value, you can delete the element from its index after identifying the index with the `indexOf()` method. If you want to use the value of the element you are removing, then use the `filter()` method, or a combination of the `indexOf()` and `splice()` methods.

Combining `indexOf()` and `splice()` Methods

Pass the value of the element you wish to remove from your array into the `indexOf()` method to return the index of the element that matches that value in the array. Then make use of the `splice()` method to remove the element at the returned index.

```
const myArray = [1, 2, 3, 4, 5];

const index = myArray.indexOf(2);

const x = myArray.splice(index, 1);

console.log(`myArray values: ${myArray}`);
console.log(`variable x value: ${x}`);
```

Output:

```
myArray values: 1,3,4,5
variable x value: 2
```

The `indexOf()` method returns the first index where the value matches the parameter passed into it, and `-1` if no value is found to match.

The `filter()` Method

The array `filter()` method takes in a function as a required argument. The function requires one parameter `currentValue`, which represents the element in the array the `filter()` method is currently on while looping through the array.

The function should perform a check that returns `true` if the `currentValue` is the value of the element you wish to remove, and `false` if not.

The `filter()` method then returns an array that contains the values from the array that match the `currentValue`.

The values the `filter()` method returns are not removed from the original array, but that functionality can be added to the function you pass into `filter()`.

```
const myArray = [1, 2, 3, 4, 5];

function removeValue(value, index, arr) {
  // If the value at the current array index matches the spec
  if (value === 2) {
    // Removes the value from the original array
    arr.splice(index, 1);
    return true;
  }
  return false;
}

// Pass the removeValue function into the filter function to re
const x = myArray.filter(removeValue);

console.log(`myArray values: ${myArray}`);
console.log(`variable x value: ${x}`);
```

Output:

```
myArray values: 1,3,4,5
variable x value: 2
```

Further Reading

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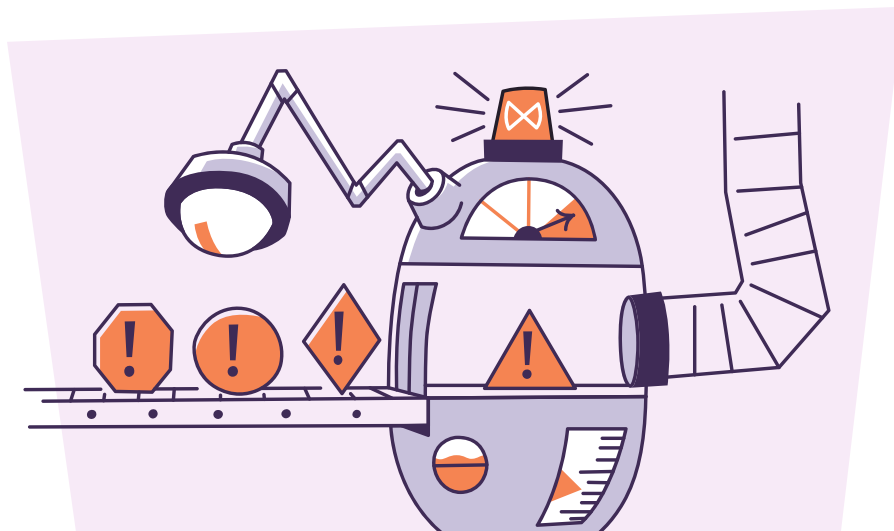
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